Sofia d'Atri

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EDUCATION

Master in Computer Science | **ETH Zurich** | Zurich, Switzerland — Sep. 2024 - Present

- Major: Visual and Interactive Computing
- Project: **Polar Collapse** (2026)
  - Developed a dungeon crawler roguelike game using **C#** and **MonoGame** in a team of 6 people.
- Project: **Dance Dance VR** (2025)
  - Created a VR rhythm game for **Meta Quest 3** in a group of 4 people, using the **Unity Engine**, **OpenXR** and **C#**.
  - The game employs **TADA**-generated meshes, **SMPL-X** parametric models and **AIST++** motion data.

Bachelor in Computer Science | **Università della Svizzera Italiana** | Lugano, Switzerland — Sep. 2021 - June 2024

- Received full tuition **scholarship** of merit.
- Bachelor thesis: **Computational Free Will - A Strange Loop In The Quantum Turing Machine**
- Project: **Raytracer** (2023)
  - Created a stochastic raytracer in **C++** with Perlin noise textures, image texturing, optimization, custom obj loader.
- Project: **JourneyTales** (2023)
  - **Full-stack** dev on a team of 12 to build a travel social app (auth, friends, real-time notifications) with **Java Spring Boot**.
- Project: **HandShake** (2022)
  - **Full-stack** dev on a team of 5 to build an instant messaging web app using the **MEVN stack**.

EXPERIENCE

Physics Simulation Scientific Internship | IST Austria | Vienna, Austria — July 2024 - Sep. 2024

- Developed a 2D **surface tracker** using the **material point method** (MPM) in Python and C++, focusing on how attributes are transferred as the mesh deforms and as the surface undergoes topological changes.

Data Management and Workflow Automation | Kellerhals Carrard | Lugano, Switzerland — Dec. 2022 - Present

- Designed automation pipelines using Microsoft Power Automate, reducing processes from a month to a few days.
- Managed databases, implemented relational models and SQL queries.

IT SKILLS

C# (OpenXR, Unity Engine), Python, C++ (Eigen, glm), Java (Spring Boot, JUnit), MEVN stack, Power BI, GitHub Actions

EXTRACURRICULAR ACTIVITIES

Head of Marketing | VIScon (VIS ETH) — Feb. 2025 - Present

- Coordinated **marketing logistics** and project management for VIScon, ETH Zurich's largest student-run tech conference.
- Managed VIScon's social media accounts, increasing **event attendance by over 40%** and content retention **by 320%**.

Social Media Management | Bequantum — Nov. 2022 - Mar. 2024

- Created posts for BeQuantum's social media profiles and managed a community of **300+ members**.

Quantum Computing Lectures and Conferences — Feb. 2022 - Mar. 2023

- Taught **quantum computing** lectures and programming labs to a total **150+ attendees** in different high schools in Italy.

LANGUAGES

Italian (native), **English** (C1), **Spanish** (intermediate), **French** (A2), **German** (beginner)

HOBBIES

Playing the piano at concerts, learning languages, active participation in student associations and event organization.